

Rafael BACHMANN

Embedded/Systems Software Developer |
Strictness and rigor enable freedom and experimentation



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SKILLS

Systems Programming	Rust, others
Embedded Technologies	STM32, embassy, KiCad, communication protocols
Fullstack Development	dioxus, axum, sqlx
Native UI/Game/Simulation Development	iced, bevy
LLM Coding	Claude, OpenCode (Big Pickle, GLMx.y, Gemini)
Electronics Tools & Instruments	Oscilloscope, Protocol Analyzer, STLinkV3, J-Link, THT and SMT soldering

EXPERIENCE

June 2023	Rust Software Developer at Paessler, Paessler, Nuremberg Paessler enables monitoring of technical infrastructure, data centers, websites, and much more. Our team (2 to 5 people), in collaboration with our customers, shipped several multi-platform Rust components. Among them were Rust modules (DLL, NATS, HTTP) of the core product, the standalone Rust multi-platform probe, C++ module integration within it, a Rust-based NATS management tool, and various Rust-based native user interfaces. Besides development, my work included training customers and colleagues as well as long-term (rather political) change processes (product development through product simplification, deep architectural changes, process improvements, and the introduction of LLM agentic coding). In an environment that eventually became increasingly stressful and alienating, our department kept up its focused, friendly, and collaborative way of working together. Rust async/await, tokio AI Ambassador Security Guardian Group Inofficial Staff Role Foreign-Function Interfacing Network communication (NATS to QUIC) Public Speaking
2022-2024	(Side Job) Rust Firmware Developer at Masetec, Masetec, Nuremberg Masetec offers a huge product range in the area of magnetic switches (reed switches) and more. Together with a very experienced specialist, I developed a small measuring device there which measures various properties of reed switches. The device has complex electronics to carry out the measurement process, a microcontroller (RP2040) to control it, and a small display to show results. It can also be connected to a PC via USB serial, where a program for displaying, configuring, and evaluating the data runs. Both the dual-core firmware and the graphical interface are written in Rust. bare metal embedded async/await, embassy, tokio tauri, js/html usb-serial

- April 2021** | **Rust Software Developer at ESRLabs, [📍](#), Nuremberg**
Development of applications for embedded Linux systems in Rust which cooperate asynchronously over the network. Due to target-platform constraints, special focus on low resource usage, e.g. multi-message UDP and single-threaded async. Focus also on reliability in distributed systems, implemented with actors, channels, and the primitives of the tokio library.
- Routing, filtering and forwarding of diagnostic data
 - Monitoring the system for suspicious activity across various areas
 - Management of encrypted information exchange between many participants over various protocols
 - Code review + pair programming + training in the Rust (and C++) ecosystem
 - Unit tests + integration tests in the Rust ecosystem (mocking, property-based testing)
 - Talks, info sessions and conference talks to advance Rust within the company
- [async/await, tokio](#) [embedded Linux](#) [Property-Based Testing](#) [Network communication](#) [Continuous Integration](#)
[Public Speaking](#)
- May 2019** | **Embedded Software Developer at MEN Mikro Elektronik Nuremberg / duagon Germany, [📍](#), Nuremberg**
Software development and testing, among others in the area of shelf/board management control and high-performance / small-form-factor I/O solutions for the rail sector.
- Development of applications for ST Microelectronics Cortex-M microcontrollers (F042/F072, G474) with a focus on availability, concurrency, and reusability.
 - Maintenance and development of cross-platform driver software (e.g. MDIS driver package, primarily Linux-based) and middleware components.
 - Active in an internal project to achieve, maintain, and uphold open-source compliance.
- [STM32](#) [C](#) [C++](#) [CMake](#) [Continuous Integration](#) [Open Source Licensing](#)

EDUCATION

- 2015 - 2019** | Studies in Mobile Embedded Systems (B.Sc.) at the [📍 University of Passau](#). Graduated in March 2019 with a thesis on formal verification and static analysis of software in embedded systems (grade 1.3). Overall grade: 1.9.
- 2014** | Started studying Computer Science (B.Sc.) at the [📍 University of Passau](#)
- 2014** | Abitur (German high-school diploma, final grade 2.1) at the [📍 Saxon State Grammar School of Music Dresden “Carl Maria von Weber”](#), with advanced courses in Mathematics and Music